TraceSmith — Cross-platform GPU Profiling & Replay

github.com/chenxingqiang/TraceSmith · C++ core + pybind11 · PyPI: tracesmith

1. Workloads / producers

PyTorch / Kineto
kernel + runtime traces

TensorFlow / XLA
graph + device events

hooks

Custom CUDA / HIP
driver / runtime APIs

Multi-GPU cluster
NCCL collectives

2. User interface

CLI
profile / record / view
export / analyze / replay

Python API
pybind11 bindings
in-process capture

Distribution
pip install tracesmith
Apache-2.0

3. Capture backends (platform-specific, lock-free ingest)

NVIDIA CUPTI
nsys / kernel / memcpy

AMD ROCm
rocprof / HIP events

Apple Metal
xctrace / Instruments

MetaX MACA
MCPTI

Huawei Ascend
CANN / msprof

Linux eBPF
host syscall / driver

events

4. Core pipeline

SPSC ring buffer
lock-free ingest
<1% overhead

Common / stacks
event timeline
XRay / host stacks

export

SBT format
string interning
delta timestamps

State / timeline
GPU state machine
stream reconstruction

5. Replay, export, debug

Replay engine
full / partial / dry-run
stream scheduler
determinism checker

Perfetto
protobuf export
~85% smaller than JSON

Tracy profiler
live timeline view

GDB RSP
GPU breakpoints
trace-driven debug

Cluster sync
NTP / PTP / CUDA
NCCL tracking

Flow: workload → CLI/Python → platform backend → lock-free ring → SBT/timeline → replay or Perfetto/Tracy/GDB.